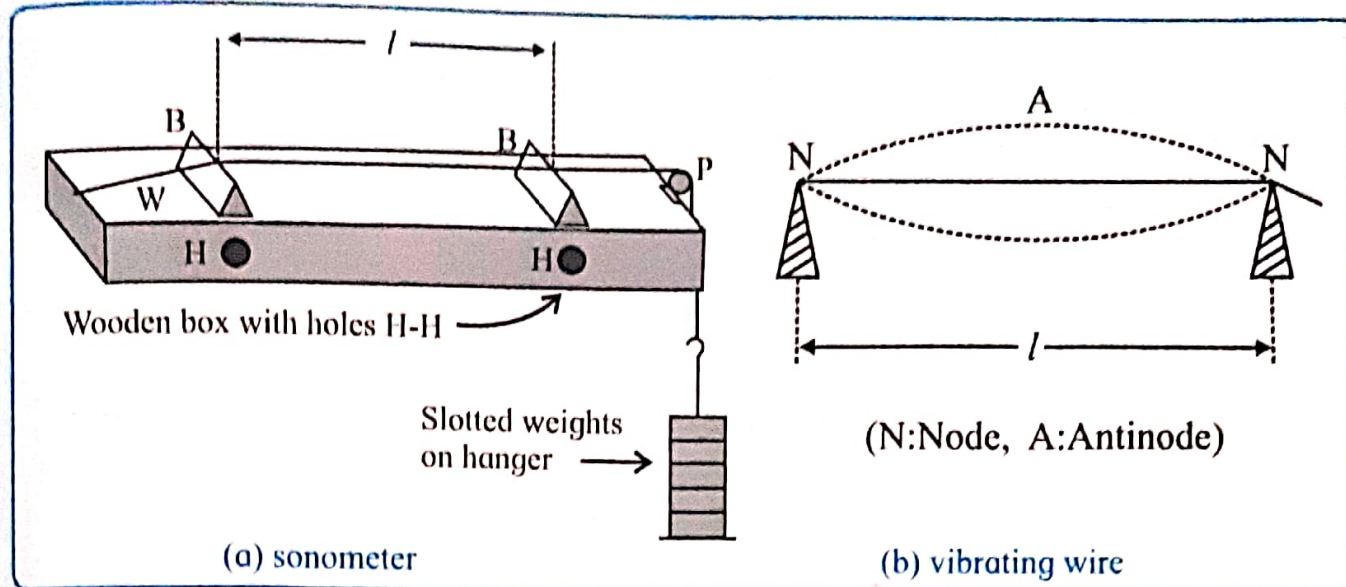


EXPERIMENT NO. 5

SONOMETER - LAW OF TENSION

Aim: To study the relation between the length of a given wire and tension for constant frequency.

Apparatus : Sonometer, slotted weights with hanger, a tuning fork of known frequency, rubber pad, light paper rider, meter scale.



Formula :

$$n = \frac{1}{2\ell} \sqrt{\frac{T}{m}}$$

n = frequency of vibrations of the wire.

T = tension applied to the wire.

m = mass per unit length of the wire

ℓ = vibrating length of the wire

$$T = 4n^2\ell^2m$$

i. e. $T \propto \ell^2$ when n and m are kept constant

Theory:

The frequency of stationary transverse vibrations of a stretched string is directly proportional to the square root of the tension in the wire, when length of vibrating string and mass per unit length of the wire are kept constant.

$$\therefore n \propto \sqrt{T} \quad \therefore \frac{n}{\sqrt{T}} = \text{constant}$$

for verification $\ell \propto \sqrt{T}$ or $\frac{\sqrt{T}}{\ell} = \text{constant}$

Procedure :

1. Apply a weight of about 0.5 kg (500 gm) including hanger to the sonometer.
2. Place a light paper rider on the wire midway between the knife edges.
3. Strike the given tuning fork on rubber pad gently and set it into vibrations.
4. Put the stem of vibrating tuning fork gently on the sonometer box.
5. Slowly change the distance between knife edges and adjust distance between them so that the paper rider falls off.

6. Measure distance between two knife edges which is the vibrating length of wire. Take on more reading for the same tension. Find the mean value of vibrating length (ℓ).
7. Repeat the same procedure with increasing tension by 500 gm applied to the wire. Note the corresponding vibrating length.
8. Calculate $\frac{T}{\ell^2}$ in each case. Plot a graph of ℓ^2 against T .
9. Determine the unknown mass by calculation and graph.
10. Determine the linear density by calculation.

Observation : $g = 980 \text{ cm/s}^2$ Frequency of tuning fork $n = 288 \dots \text{Hz}$

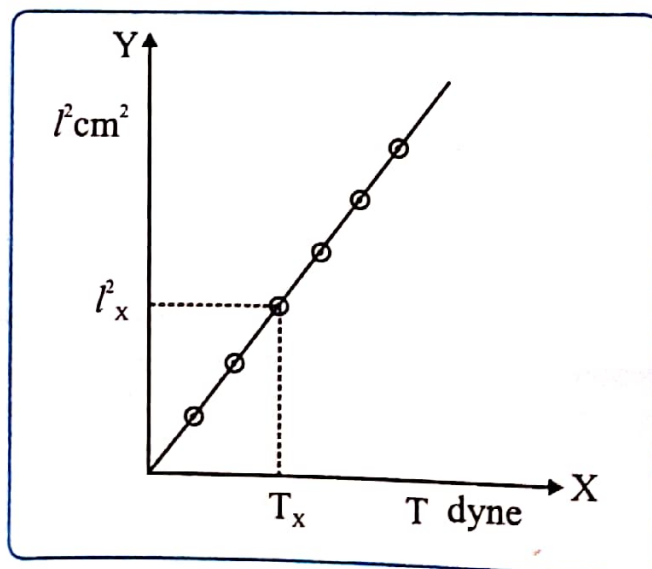
Obs. No.	Mass attached to the wire including hange Mgm	Tension $T = Mg$ dyne	Vibrating length of wire			ℓ^2 cm^2	T/ℓ^2 dyne/cm
			$\ell_1 \text{ cm}$	$\ell_2 \text{ cm}$	Mean $\ell \text{ cm}$		
1	500	490×10^3	10	11.5	10.75	115.56	4240.22
2	1000	980×10^3	12.5	14	13.25	175.56	5582.13
3	1500	1470×10^3	16	17.5	16.75	280.56	5239.52
4	2000	1960×10^3	19	20.5	19.75	390.06	5023.58
5	Unknown	$T_x = 1250 \times 10^3$			$\ell_x = 15.125$		

Calculations :

Ob. No.	1	2	3	4	5
$\log T = a$	5.6989	5.9912	6.1674	6.2923	
$\log \ell^2 = b$	2.6028	2.2444	2.4480	2.5911	
$a - b = c$	3.0634 3.0961	3.7468	3.7194	3.7091	
Antilog $c = T/\ell^2$	4240.22	5582.13	6239.52	5023.58	

Graph : A graph of ℓ^2 against T

Scale $Y = \frac{100}{500 \times 10^3} \text{ cm}^2$
 $X = \dots \text{dyne}$



A graph of L^2 vs T

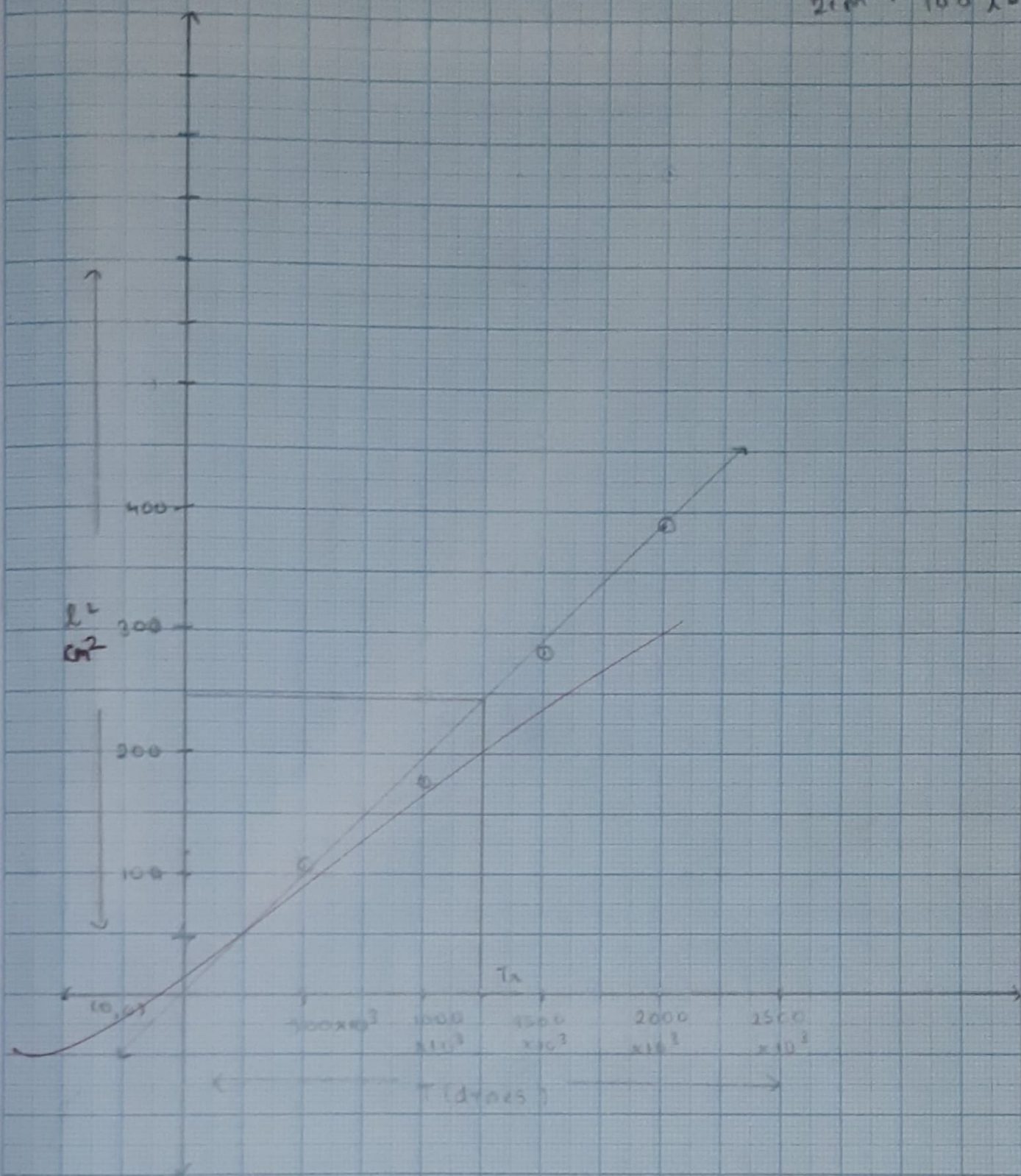
Scale :

X axis

2cm = 500×10^3

dynes

2cm = $100 \text{ } L^2$



Sign of teacher

Result :

1. As $\frac{T}{\ell^2} = \text{constant}$, when n and m are kept constant. The law of tension is verified.
2. The graph of ℓ^2 against T is straight line $\therefore T \propto \ell^2$ Hence law of tension is verified.
3. Unknown mass by calculation = 1.135g
4. Unknown mass by graph = 1.135g

Precautions :

1. Wire should be free from kinks and of uniform cross section.
2. Keep the paper rider at the mid point of knife-edges.
3. The stem of tuning fork should rest on the sonometer box.
4. Strike the tuning fork on the rubber pad gently.

Questions

1. State the second law of vibrating string.

Statement :

The fundamental frequency of a stretched string is directly proportional to the square root of its tension, provided the vibrating length and mass per unit length remain constant.

2. Why do we keep frequency constant instead of keeping vibrating length constant to verify second law of vibrating string?

In a sonometer experiment, the frequency of the tuning fork is fixed and cannot be changed. Therefore, to verify the second law of a vibrating string, the frequency is kept constant and the tension is varied by adjusting the load.

3. How many nodes and antinodes are formed when the wire vibrates in fundamental mode?

When the wire vibrates in the fundamental mode, two nodes are formed at the ends of the wire and one antinode is formed at the centre.

Remark and sign of teacher :